

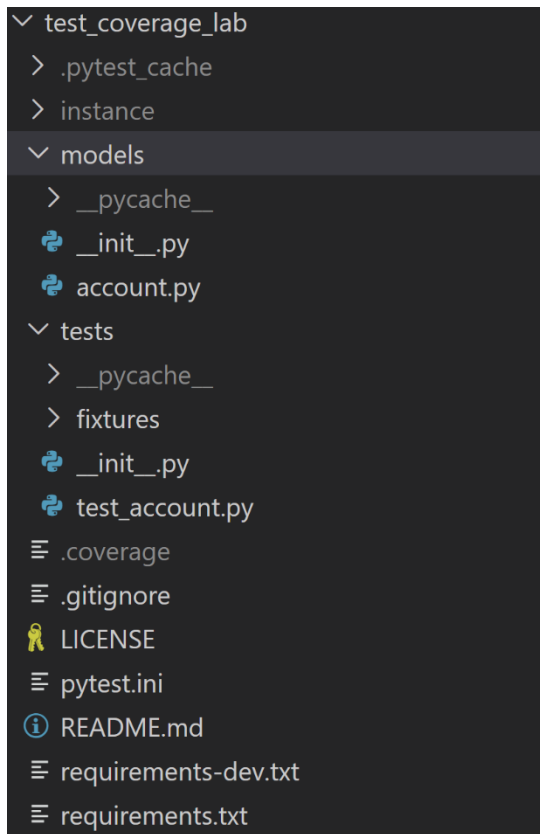
Testing and CI Lab Report

Phase 1 Report: Test Coverage Lab

Link to forked repo: <https://github.com/johns216/Group-2>

Reviewed Ethan Guillem (ethanvfour) and reviewed by Peter Nguyen (Peekans)

Folder from github repo cloned to local machine:



Initial pytest run for Test Coverage Lab:

```
===== tests coverage =====
coverage: platform win32, python 3.14.0-final-0
Name                               Stmts  Miss  Cover   Missing
-----
models\__init__.py                  7      0   100%
models\account.py                  58     11    81%   30, 47-49, 60-62, 67, 73, 102-103
tests\__init__.py                   0      0   100%
tests\test_account.py              78      0   100%
-----
TOTAL                             143     11    92%
===== 9 passed in 1.20s =====
```

Link to setup of test coverage lab issue and PR:

Issue: <https://github.com/kenner1-unlv/Group-2/issues/43>

PR: <https://github.com/kenner1-unlv/Group-2/pull/44>

Link to my issue and acceptance criteria: <https://github.com/kenner1-unlv/Group-2/issues/83>

Link to related PR for this issue: <https://github.com/kenner1-unlv/Group-2/pull/94>

Coverage and individual test passing for implemented delete account test:

```
tests/test_account.py::test_delete_account PASSED
```

```
===== tests coverage =====
coverage: platform win32, python 3.14.0-final-0

Name                Stmts   Miss  Cover   Missing
-----
models\__init__.py      7      0   100%
models\account.py     58      9    84%   30, 47-49, 60-62, 67, 73
tests\__init__.py       0      0   100%
tests\test_account.py  84      0   100%
-----
TOTAL                  149      9    94%
===== 10 passed in 1.10s =====
```

This test tests whether an account can be successfully deleted from the database. It tests a setup account and email and deletes the email. The test then verifies whether the deleted account is in the Accounts database. If not, the test passes. If it is still, the test does not pass.

Link to Ethan's (ethanvfour) PR for Test coverage I reviewed: <https://github.com/kenner1-unlv/Group-2/pull/78>

Ethan's PR merged and issue was closed. All tests accounted for and setup presented clearly.

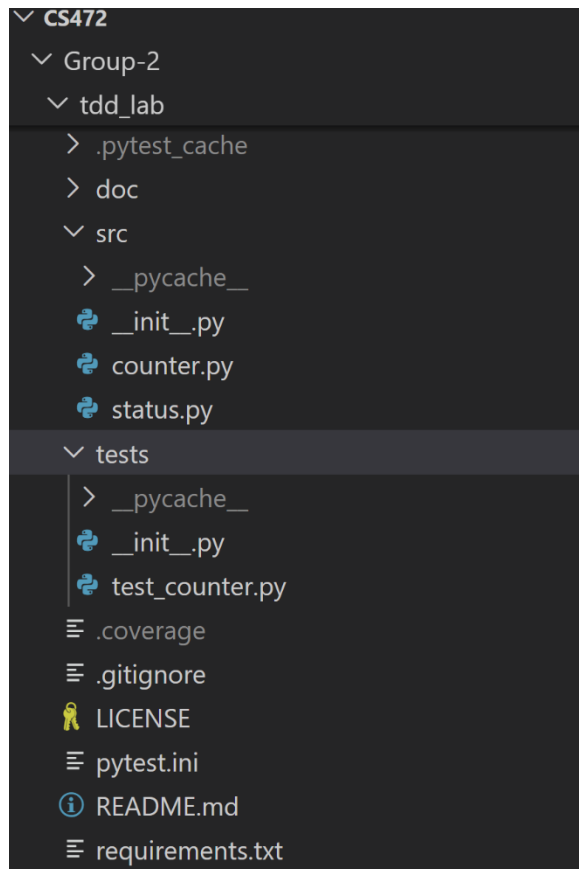
Feedback for personal Test coverage PR by Peter (Peekans):

Everything worked properly. One issue raised in the review was that an edge case for determining if an account exists could be added as well, but it wasn't required as part of the test, so I chose not to implement extra.

Link to approved and closed PR: <https://github.com/kenner1-unlv/Group-2/pull/94>

Phase 2 Test-Driven Development Lab

Setup screenshot of TDD lab folder on local machine:



Initial setup pytest screenshot running:

```
===== tests coverage =====
coverage: platform win32, python 3.14.0-final-0
Name                Stmts  Miss  Cover   Missing
-----
src\__init__.py       3      0   100%
src\counter.py       36      1    97%   33
src\status.py         7      0   100%
-----
TOTAL                  46      1    98%
===== 7 passed in 0.32s =====
```

Link to TDD setup issue: <https://github.com/kenner1-unlv/Group-2/issues/46>

Link to TDD setup PR: <https://github.com/kenner1-unlv/Group-2/pull/47>

Link to personal issue and acceptance criteria: <https://github.com/kenner1-unlv/Group-2/issues/96>

Link to individual TDD PR including test case: <https://github.com/kenner1-unlv/Group-2/pull/97>

This feature tests handling unsupported HTTP methods by returning an error code 405 in the database. If there is an unsupported HTTP method listed, the test will fail. If there is no unsupported HTTP method, the test will pass.

Evidence of test results:

Initial failure of test (before endpoint was written):

```
===== tests coverage =====
coverage: platform win32, python 3.14.0-final-0
Name           Stmts  Miss  Cover   Missing
-----
src\__init__.py    3     0   100%
src\counter.py    38     1    97%    33
src\status.py      7     0   100%
-----
TOTAL             48     1    98%
===== short test summary info =====
FAILED tests/test_counter.py::TestCounterEndpoints::test_handle_invalid_http_methods - assert None is
not None
===== 1 failed, 8 passed in 0.87s =====
```

After implemented test and endpoint:

```
tests/test_counter.py::TestCounterEndpoints::test_handle_invalid_http_methods PASSED

===== tests coverage =====
coverage: platform win32, python 3.14.0-final-0
Name           Stmts  Miss  Cover   Missing
-----
src\__init__.py    3     0   100%
src\counter.py    41     1    98%    33
src\status.py      7     0   100%
-----
TOTAL             51     1    98%
===== 9 passed in 0.55s =====
```

Peter Nguyen (Peekans) once again reviewed my PR. He left great feedback. Initially my red test had passed but had less tests overall, which did not indicate a failure state. That was adjusted so there was a clear failure case followed by a success case. He also noted there was a version mismatch before, which I didn't catch. That was also adjusted to be current and the tests were

rerun. He suggested reverting some changes in the counters.py file but they were necessary endpoints for my test to complete and validate so they were approved.

Link to Ethan's (ethanvfour) PR for TDD I reviewed: <https://github.com/kenner1-unlv/Group-2/pull/80>

Everything in Ethan's PR worked perfectly. All acceptance criteria were addressed, the PR was linked to the relevant issue, and it was closed. Red test failed initially and green test passed afterwards. Slight refactor also implemented well.

Link to approved personal PR for TDD Lab: <https://github.com/kenner1-unlv/Group-2/pull/97>

Phase 3 Continuous Integration Lab:

Link to my fork: <https://github.com/johns216/Group-2>

Evidence of sync with upstream repo:



Evidence that ci_lab starter workflow ran successfully:

```
===== tests coverage =====
coverage: platform win32, python 3.14.0-final-0
Name           Stmts  Miss  Cover   Missing
-----
src\__init__.py    3     0   100%
src\counter.py   87     5    94%   71, 85, 96, 99-100
src\status.py     7     0   100%
-----
TOTAL             97     5    95%
===== 22 passed in 0.67s =====
```

Link to PR issue and its acceptance criteria: <https://github.com/kenner1-unlv/Group-2/issues/99>

Link to PR: <https://github.com/kenner1-unlv/Group-2/pull/102>

Workflow evidence:

Lint
failed 11 minutes ago in 23s

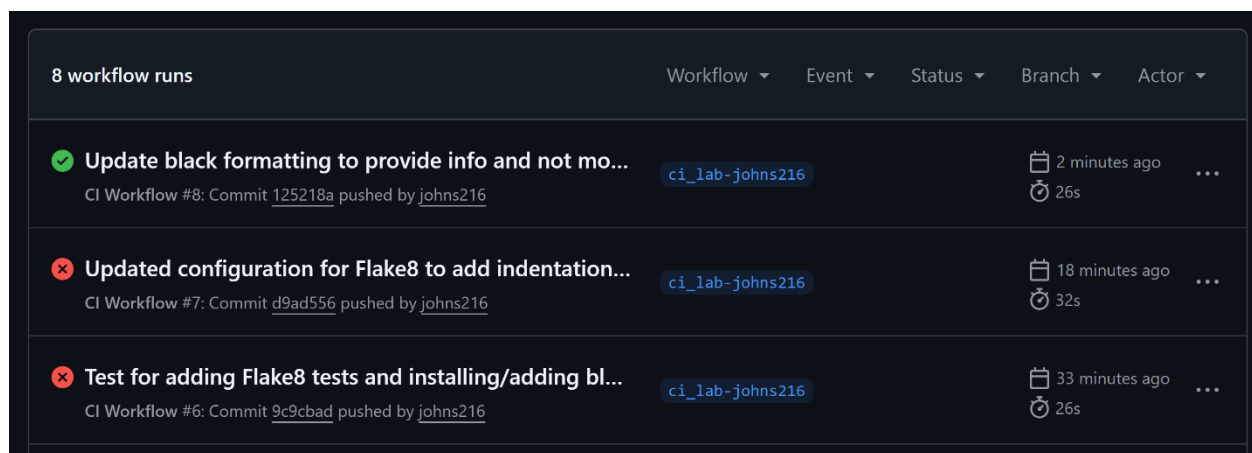
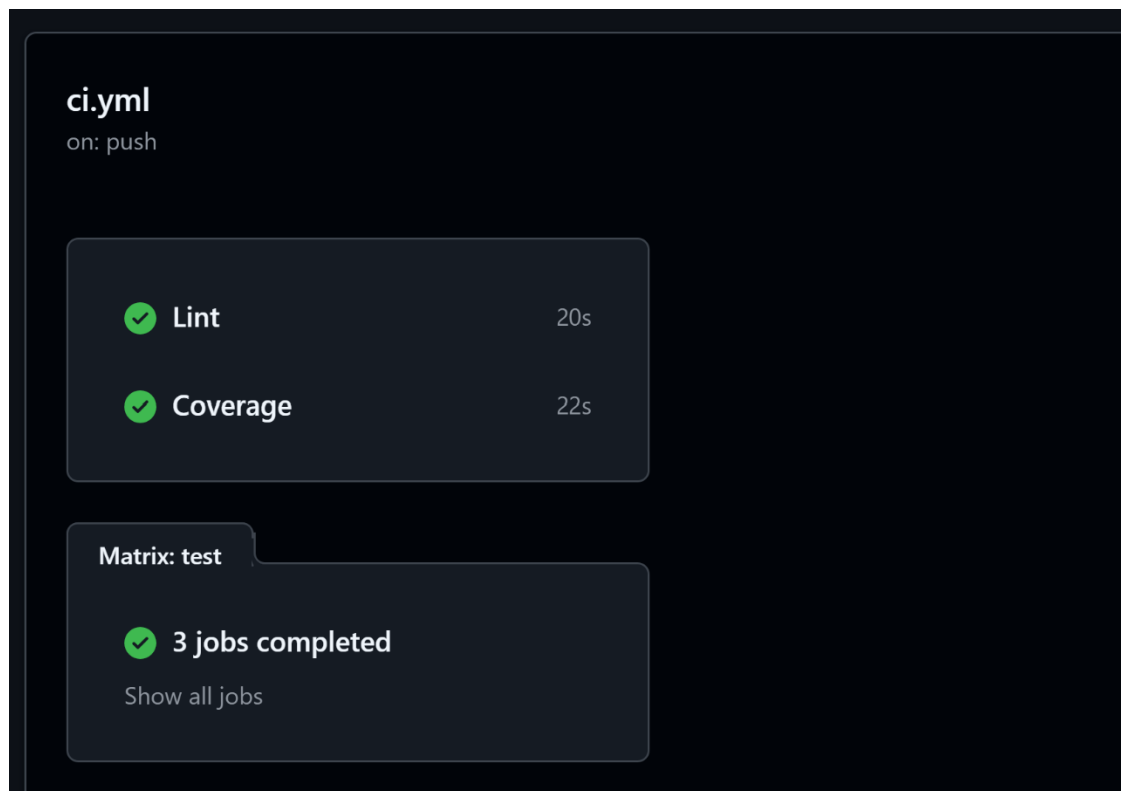
Search logs

- > Set up job
- > Checkout Repository
- > Set Up Python
- > Cache Python Dependencies
- > Install Dependencies
- > Run Flake8 Linting

```
111 HTTP_409_CONFLICT = 409 ^
112 15 E302 expected 2 blank lines, found 1
113 17 E501 line too long (82 > 79 characters)
114 1 W292 no newline at end of file
115 1 W391 blank line at end of file
116 34
117 Error: Process completed with exit code 1.
```

Run Black Formatting Check

```
1 ▶Run black --check src
11 would reformat /home/runner/work/Group-2/Group-2/ci_lab/src/status.py
12 would reformat /home/runner/work/Group-2/Group-2/ci_lab/src/counter.py
13
14 Oh no! ✖️💔✖️
15 2 files would be reformatted, 1 file would be left unchanged.
16 Error: Process completed with exit code 1.
```



Link to Flake8 test controlled failure github actions: <https://github.com/johns216/Group-2/actions/runs/35391966284>

Link to Black formatting controlled failure github actions: <https://github.com/johns216/Group-2/actions/runs/35393257922>

Link to merged commits: <https://github.com/kenner1-unlv/Group-2/pull/102/commits>

Link to successful CI run: <https://github.com/johns216/Group-2/actions/runs/35394680971>

Link to Ethan's (ethanvfour) PR I reviewed for Phase 3 CI Lab: <https://github.com/kenner1-unlv/Group-2/pull/82>

I examined Ethan's CI logs and test results for his PR listed above. He added a controlled failure to change the workflow and CI logs. After, he analyzed the root cause and fixed the controlled failure so all tests passed successfully. PR was approved and merged, as all qualifications for his acceptance criteria listed in the relevant issue was met.

Reflection:

Upon doing this lab, I learned a lot about workflow and testing. It was useful to be able to write sample tests and perform them to analyze our code. Pytest is incredibly helpful for that. Also, learning about Continuous Integration is necessary to keep repos on track with all the fixes and errors that can occur. I think we all got a practical understanding of raising issues, solving them, adding tests, forming pull requests, reviewing them, and merging them successfully. This was a very useful lab.